

RESEARCH ASSIGNMENT I

SECTION A

1. Database is a digital storage where data is organized to allow it for easy access and management.

Databases are essential in the modern world since they easily store, organize and process massive amounts of data from various sources and the information helps with our daily lives.

Examples

- * A hospital stores patients' medical records and appointments information in their database.
- * An online store storing customers' information like their name, surname, contact details and location in their database.

2. A data warehouse is a centralized digital storage system where multiple databases are stored.

A data warehouse uses a process called OLAP which handles complex data analysis and historical reporting whereas databases use a process called OLTP which handles real time day to day operations.

A company would use a data warehouse because it acts as a centralized storage system to store and clean data from various sources.

3. A data lake is a centralized storage folder that stores large amount of raw, unprocessed data until it is needed.

A data lake stores raw, unprocessed and unstructured data while a data warehouse stores cleaned and structured data.

A business would use a data lake because it is durable and low cost because of its ability to leverage and scale object storage.

4. A data mart is a subset of a data warehouse focused on a particular line of business, department or subject area.

It is related to data warehouses since it exists as a subset of a company's organization's larger enterprise data warehouse.

Specific business departments and teams such as marketing, sales and finance would typically use a data mart.

Structured data	Semi-structured data	Unstructured data
Has a fixed schema and fits neatly into rows and columns. example: An excel spreadsheet.	Information that does not use a rigid row and column table structure, but still use tags, keys or metadata to group elements and build a clear hierarchy. example: XML	Information that does not have a predefined format or a fixed schema. example: Email.

6. Data Modelling is the process of creating a visual representation of an information system to communicate connections between data points and structures.

Data modeling makes it easier for developers, data architects and business analysts to view and understand relationships among the data in a database or data warehouses.

Types of data models

- * Conceptual data model describes the database at a very high level and is useful to understand the needs or requirements of a database.
- * Entity relationship model is a high level data model which is used to define the data and the relationships between them.

7. Data Mining is the process of discovering meaningful patterns and insights from large datasets using statistical and machine learning techniques.

Examples

- * Online stores analyze purchase histories to see which items customers frequently buy together (e.g., finding out if people usually buy pasta and cheese).
- * Hospitals mine the patients' medical records and histories to determine early signs for conditions like diabetes.

8. Database Management System is a software that lets users and applications create, update, retrieve and secure data in a database.

DBMS Platforms

* MySQL: An open source relational database management system that stores data in structured tables.

Suitable use: Powering content management systems like WordPress and web application built on LAMP stack.

2. PostgreSQL: An advanced open source object-relational database known for strong standards compliance, extensibility and handling complex queries for large volumes of data.

Suitable use: financial systems, geographic information systems (GIS).

3. Oracle DB: A commercial, high scalable multi-model relational database management system that offers enterprise grade security, high availability and advanced data processing.

Suitable use: ~~Real time analytics and catalog systems~~. Banking and large scale corporate data environments.

4. MongoDB: A popular NoSQL document database that stores data in flexible, JSON like documents / making it easy to change data structures over time.

Suitable use: Real time analytics and catalog systems.

9. ETL (Extract, Transform, Load) is a data integration process that combines, cleans and organizes data from multiple sources.

ETL helps enterprises to gather information from fundamentally different sources and transform it into trusted data that can be used for analytics in different environments, from on-premise servers to cloud platforms.

Extract: This step collects data from various source systems, including databases, spreadsheets, cloud based platforms and files. The data is in its raw form, usually in large batches or portions.

Transform: The data is transformed to ensure it is clean, consistent and usable. Transformation includes tasks such as data cleaning, removing duplicated data, data validation and authentication and summarizing raw data in standard formats for consistency.

Load: The process loads it into a target system, such as a data warehouse or a database optimized for analytical queries.

10. CSV: A simple, text-based, row oriented format where values are separated by commas, it has no strict built-in schema or data types.

Use: Use CSV when you need human readable data.

JSON: A light-weight, semi structured, text based format that uses key value pairs and supports

SOURCES SECTION A

- * IBM. (2025). What is a Data Warehouse? IBM Think
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- * Database Systems: Design, Implementation and Management 11e by C. J. Date / Morris

hierarchical data.

Usage: for web applications APIs.

SECTION B

11. Artificial Intelligence is a branch of computer science that creates machines capable of simulating human intelligence, such as learning, reasoning, problem solving and perception.

Difference between Narrow AI and General AI
General AI works beyond the narrow specialization of narrow intelligence. It can understand and apply knowledge in multiple countries, adapt to new situations and learn from very limited information while Narrow AI focuses on a single task and is restricted from moving beyond that task to solve unknown problems but general AI can solve many problems using human-like intelligence.

12. Machine learning is a branch of AI that allows computer systems to learn from data and improve from experience without being explicitly programmed.

Traditional programming uses explicit human written rules and data to produce an output, whereas machine learning uses data and desired outputs to automatically discover the underlying rules.

* Supervised learning: Trains a model to predict the "correct" output for a given input.

* Unsupervised learning: Trains a model to discern intrinsic patterns, dependencies and correlations in data.

* Reinforcement learning: Trains a model to evaluate its environment and take an action that will garner the greatest reward.

13. Deep learning: A type of machine learning that relies on neural networks to learn from training data.

Artificial intelligence, machine learning and deep learning form a nested hierarchy where AI is the broad umbrella, machine learning is a subset of AI and deep learning is a specialized subset of machine learning.

A neural network is a machine learning algorithm that functions similarly to how our brain works, with neurons transmitting information between layers of interconnected nodes.

14. Natural Language Processing (NLP): A subfield of computer science and artificial intelligence that uses machine learning to enable computers to understand and communicate with human language.

Example

It can generate articles, reports, marketing copy and

creative writing based on prompts provided by users.

15. Large language Model (LLM) are a type of artificial intelligence that uses machine learning algorithms to replicate human language.

* ChatGPT (by OpenAI): A conversational AI assistant used for drafting text, answering complex questions, writing and debugging computer code and brainstorming ideas.

* Gemini (by Google): A multimodal AI model used to integrate text, code and image reasoning across applications, summarize large volumes of data and power enterprise cloud services.

16. Generative AI is a type of artificial intelligence that creates new content such as text, images, audio and code by learning statistical patterns from existing training data.

* ChatGPT: writes human like text, answers complex questions, summarizes long documents and generates code.

* Microsoft Copilot: Helps draft emails, summarize meetings and generate spreadsheets or documents.

* DALL-E 3: Creates realistic pictures, illustrations, and visual art directly from written text descriptions.

17. Training data: Information that is used to teach a machine learning model how to make predictions, recognize patterns or generate content.

The quality of training data is important because strong data quality determines whether an AI model succeeds or fails.

Biased data causes the AI to replicate and amplify historical prejudices, leading to unfair outcomes.

18. Computer vision is a branch of artificial intelligence that trains computers to extract meaning, patterns, and useful information from digital images and videos.

A camera or sensor captures a picture or video as raw visual data. The computer turns the image into a huge grid of numbers called pixels. The system adjusts contrast or removes blur to make the picture clear. Special algorithms scan the grid to spot basic lines, shapes and edges.

Examples

- * Self driving cars use cameras and sensors to track surroundings in real time.
- * Doctors use Microsoft Azure tools, powered by computer vision to scan X-rays, MRIs and CT scans.

SOURCES SECTION B

Cost of a Data Breach report 2026.

Attackers are weaponizing AI.

* Deep learning vs Machine learning : A beginners guide

* Neurons in Deep learning : The building Blocks of Neural Networks

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* [https://www.azure.microsoft.com > cloud-computing - dictionary](https://www.azure.microsoft.com/cloud-computing-dictionary) , Microsoft Azure

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SECTION C

19. Application Programming Interface (API) are mechanisms that enable two software components to communicate with each other using a set of definitions and protocols. eg, The weather bureau's software system contains daily ~~the~~ weather data. The weather app on your phone talks to this system via APIs and shows you daily weather updates on your phone.

APIs are more than just technical interfaces, they are the foundational building blocks of modern software architecture. By enabling communication between diverse systems, they foster innovation, scalability and efficiency.

20. An API key is a unique identifier used to authenticate software and systems attempting to access other software or systems through an application programming interface (API)

How they can be stored and protected

- * Combining multiple authentication methods.
- * Secure key storage.
- * Rotating or replacing keys.

A key that is exposed publicly can enable quick unauthorized access to your API and can lead to unauthorized usage, data exposure,

service abuse and reputational damage.

21. A REST API is an API that ~~confronts~~ conforms to the design principles of the representational state transfer (REST) architectural style, a style used to connect distributed hypermedia systems.

Main HTTP methods

- * GET : Retrieves data from the server. It is a read-only operation and does not change the state of the data.
- * POST : Sends new data to the server to create a brand new resource.
- * PUT : Updates or completely replaces an existing resource on the server.
- * DELETE : Permanently removes a specific resource from the server.

22. JSON is a text based format for storing and exchanging data in a way that's both human readable and machine parsable.

JSON's language independent nature makes it an ideal format for exchanging data across different programming languages and platforms hence it is commonly used with APIs.

Example of JSON object

```
json
{
  "firstName": "John",
```



```
"lastName" : "Doe"  
}
```

23. Prompt Engineering is the process of writing, refining and optimizing inputs to encourage generative AI systems to create specific high quality outputs.

It is an important skill when working with AI because clear instructions prevent AI from guessing or providing vague, generic answers.

Examples

Poor prompt: Write an email to my boss saying I am sick

Improved prompt: Act as a professional employee, Write a short, polite email to my manager, Marie, letting her know I have a fever and cannot come into the office today. Mention that I will check urgent emails by noon if I feel better, keep it short

Prompt that's poor: Explain quantum computing

Improved prompt: Explain quantum computing to a 10 year old using a simple everyday analogy, limit the response to 3 short paragraphs and avoid heavy technical jargon.

24. A MAP function is a built-in programming tool that transforms data by applying a specific operation to every item in a collection and returning a new collection with the results.

What it does

It takes an input collection, then takes a temporary or predefined rule and executes it on the first item, then second, third and so on. It then creates a new collection of transformed pieces of data.

Example

#1. Define the original list of numbers

numbers = [1, 2, 3, 4, 5]

#2. Use map() with a lambda

We wrap it in a list() to convert the map back into a readable list.

doubled_numbers = list(map(lambda x: x*2, numbers))

#3. Print the result

print(doubled_numbers)

output: [2, 4, 6, 8, 10]

25. Cloud computing is the delivery of on-demand computing services, including data storage, servers, databases, networking and software.

Cloud service models

* Infrastructure as a service (IaaS): The most flexible model where a provider rents out foundational IT building blocks like virtual

servers, storage, and networking over the internet. You manage the OS, system, data and applications while the provider manages the physical hardware.

example: Amazon EC2 (Elastic Compute Cloud)

* Platform as a Service (PaaS): A model that supplies a ready-to-use, pre configured environment. It includes OS, databases, and development tools so developers can build, test and deploy applications without worrying about managing the underlying hardware or servers.

Example: Google App Engine.

* Software as a Service (SaaS): A fully ~~formatted~~ finished software application delivered over the internet via a web browser on a subscription basis.

Example: Microsoft 365

26. Version Control System is a software tool that tracks and manages changes to source code, or files over time.

Git is a free, open-source, distributed version control system originally developed by Linus Torvalds in 2005.

Analysts and developers can try out complex algorithms, new features or exploratory data models without messing up the working production code.

*Commit: A saved snapshot of your project's files at a specific moment in time.

*Branch: An independent line of development that diverges from the main project.

27. A Jupyter Notebook is a free open-source web application used to create and share interactive documents that combine live programming code, computational outputs, data visualizations and exploratory text in a single workspace.

It is commonly used with Python.

It is common amongst data analysts and scientists because they use it for cell based execution, inline visualizations and multimodal data storytelling.

28. Python is a high level, general purpose programming language known for its clean syntax, code readability and beginner friendly design.

Python Libraries

- * Pandas : Used for data manipulation and analysis.
- * Numpy : Used for numerical computing.
- * Matplotlib : Used for data visualization.
- * Skit-learn : Used for machine learning.

SOURCES SECTION C

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IBM Think
- * <https://www.oracle.com> > africa > database > what-is-
json , Oracle
- * <https://www.com> > Think > Topics : IBM Think.
- * Geeks 4 Geeks. (2026). Python Programming Language.

SECTION D

30. Data Ethics is the moral obligations of collecting, storing and using data.

Examples

* Unconsented Facial Recognition & tracking

A retail chain installs hidden facial recognition cameras in its stores to track shoppers' emotional response to products and build secret behavioural profiles. The data is collected without the shoppers' knowledge.

* Predatory Surveillance Advertising

An e-commerce platform tracks a user's browsing history to detect signs of financial distress or mental health struggles.

* Algorithms bias in Hiring

An AI recruitment tool trained on historical hiring data from a male-dominated tech company penalizes resumes containing the word "women's". This automates historical gender discrimination.

31. Data governance: The people, processes and technologies used to manage, protect and use data. It ensures consistent and trustworthy data that organizations can leverage as an organizational asset.

key components of a good data governance framework

* Rules: The policies, requirements, standards and controls, and the rules that govern how different people will enforce them.

* People: The stakeholders and stewards that make and enforce the rules.

* Processes: Governs the data while creating value, managing complexity and cost, and ensuring compliance.

32. Bias in AI is the discrimination embedded in AI systems, resulting in unfair or harmful results.

For an AI model to be biased, it means its internal logic or outputs systematically favor certain groups of people over others based on traits like race, gender and age.

Example.

In a landmark 2018 study by researcher Joy Buolamwini, commercial Facial Recognition Systems built by major technology companies were found to misidentify darker skinned women at rates as 35%, compared to error rates under 1% for lighter skinned men.

The AI was biased because its training data lacked adequate representation of darker skin tones and diverse facial features.

33.

33.

Role	Focus/Role	Skills	Tools
Data Analyst	Interpret historical data to create reports, dashboards and answer business questions.	SQL, basic statistics, data visualization, business communication	Excel, PowerBI, Tableau, SQL
Data Scientist	Forecast future trends and build machine learning/statistical models	Advanced Statistics, machine learning algorithms, Python/R, data wrangling	Python, R, Scikitlearn, Pandas, Jupyter.
Data Engineer	Build and manage data infrastructure, pipelines and storage systems.	Software engineering, database management, data warehousing, ETL	SQL, Apache Spark, kafka, Snowflake, Airflow, Git.
AI Engineer	Build, optimize and deploy AI driven applications and RAG pipelines.	Deep learning, system architecture, API integration, vector database.	Python, Pytorch, TensorFlow, LangChain, Vector DBs.

334. Business Intelligence is a set of software tools and processes used to collect, organize and analyze company data so leaders can make better decisions.

BI tools

* Microsoft Power BI : A cloud based and desktop business analytics service by Microsoft that connects to a vast range of data sources, allowing users to create live dashboards and interactive reports.

* Salesforce Tableau : A powerful visual analytics platform focused on deep data visualization, letting users easily build interactive, drag and drop data stories and visual graphs.

SOURCES SECTION D

- * IBM. (2025). Data privacy Guide to AI and Machine learning. IBM Think
- * IBM. (2023). Why Data Governance is Essential for Enterprise AI. IBM Think.
- * ALX. (2026). Data Analytics.

SECTION 2

35. The concept I found most interesting was Machine learning. It shows how computers can learn patterns from data and use those patterns to make decisions. ~~Also~~
36. The concept I found most difficult to understand was APIs, especially how different applications communicate with each other using an API. I understand the basic idea but needed more explanation to understand how requests and responses work.
37. As a Computer Science and Statistics graduate, I believe data and AI will have an important role in my future career. I can use data analysis to identify patterns, understand trends and support decision-making.